

Fed-MDBSCAN-G: An Empirical Study of Filtering Failures and Decision Pathways in Heterogeneous Federated Learning

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Abstract—Geometric separation of federated updates can reflect legitimate data heterogeneity as well as poisoning. We evaluate this ambiguity in Fed-MDBSCAN-G, an implemented defense combining a geometric-median acceptance region, relative-density partitioning, shared-neighbor cluster validation, and temporal alarm memory. A canonical evaluation across four datasets comprises 2,130 planned method-condition-seed runs, with 2,125 valid outcomes and five retained failures. Under attack-free extreme heterogeneity, the full method rejects 24.37–42.57 percent of honest submissions; each dataset records alarms in all 90 evaluated clean rounds. Across 36 paired ablation endpoints, its mean final-accuracy advantage over the initial filter alone is 0.081 percentage points, with differences in both directions. Dataset-level mean attack success exceeds 96 percent in the canonical patch-backdoor block. A separate exploratory shared-checkpoint study identifies a decision pathway: in one selected MNIST checkpoint, changing the batch regime reduces rejection from 34 to 18 of 90 honest clients. Matched repetitions show that ten clients avoid additional rejection by remaining outside the natural clusters subjected to consensus validation. This explains their decisions in the local implementation without establishing long-horizon benefit or attack resistance. Additional direction and temporal-score probes have distinct limitations and do not establish a common failure mechanism. The evidence supports a bounded assessment of this implementation; it does not establish failure of geometric defenses as a class.

Index Terms—Federated learning, poisoning, robust aggregation, data heterogeneity, relative density, backdoors, reproducibility.

I. INTRODUCTION

FEDERATED learning aggregates locally computed updates while retaining training examples at their originating clients [1]. This separation limits direct inspection of local training and creates opportunities for poisoned data or manipulated updates [2], [3]. A server can examine update geometry, but distance from the majority need not identify malicious intent: clients holding different classes or different numbers of examples can also submit distinct updates.

We study whether a multi-density filter can resolve this ambiguity. MDBSCAN separates density regimes and recovers low-density natural clusters in centralized clustering [4]. Our adaptation, Fed-MDBSCAN-G, combines these ideas with a geometric-median acceptance region and cluster-level consensus checks. The design aims to preserve useful heterogeneous updates while removing harmful deviations. These objectives must be tested independently: a geometric rejection rule does

not establish attack attribution, and low attack success is uninformative when the model itself fails to learn.

The evaluation consequently separates four questions. Does filtering improve final accuracy relative to simple aggregation? Does it reject honest clients under attack-free heterogeneity? Does the alarm distinguish attacked from clean rounds? Do additional density and temporal components contribute beyond the initial geometric filter? Matched attack-disabled controls and targeted backdoor measurements address damage and security outcomes separately.

The contribution has three parts with different evidential scope. First, the canonical evaluation quantifies utility, honest-client exclusion, alarm errors, and targeted attack success while retaining failed runs. Paired ablations describe the contribution of the upper layers in the tested block; a small mean difference is not an equivalence result. These observations constrain the claims that can be made for this implementation.

Second, shared-checkpoint diagnostics trace a selected honest-client retention contrast to the local decision pipeline. Some low-density updates avoid additional rejection because they remain outside the natural clusters subjected to consensus validation. Matched fixed-matrix attack replays distinguish a Min-Max pathway outside those clusters from patch updates that enter clusters and pass the test. These are explanations at recorded states, not evaluations of retrained defenses or independent validation of generality.

Third, exploratory checks document different limitations of candidate neighbor-direction, clean-root, and temporal-norm scores. Synthetic specificity controls, fixed-matrix direction changes, and new training runs answer different questions; their evidence is separated in Table VI. Only the profile-aware temporal intervention measures training damage after modifying that attack parameter. We claim neither a common impossibility mechanism nor general superiority, and do not transfer baseline convergence or security guarantees to the composite filter.

II. RELATED WORK AND SCOPE

A. Robust Aggregation and Trust Assumptions

Statistical robust aggregation includes neighbor-based selection such as Krum [5], coordinate-wise robust estimators [6], and geometric-median aggregation [7]. FLTrust introduces a server-side trusted dataset and similarity-based weighting [8]. These approaches impose different information and modeling assumptions. Their theoretical properties cannot be transferred

to a composite filter merely because it uses a related distance statistic.

B. Cluster Structure and Security Decisions

Clustering-based defenses inspect structure among submitted updates. FLAME combines HDBSCAN-based selection with clipping and noise [9]; it should not be characterized as an ordinary fixed-radius DBSCAN baseline. MDBSCAN instead addresses multiple density regimes through relative-density partitioning and shared-neighbor clustering [4]. We reuse this motivation without assuming that natural clusters correspond to honest participants. Coordinated attackers may themselves form coherent groups.

The adaptation is not a complete execution of the original MDBSCAN pipeline. In Qian et al. [4], Algorithm 1 uses nearest-neighbor sets without the additional radius cutoff used here; Algorithm 2 applies DBSCAN to points outside the recovered natural clusters and assigns its noise points to the nearest clusters. Our defense instead adds a 3ϵ cutoff and a minimum natural-cluster size of two, and replaces that continuation with geometric acceptance and cluster-consensus rejection. These are implementation choices whose effects must be evaluated separately.

Our checkpoint result concerns how the local implementation treats low-density points outside recovered natural clusters. MDBSCAN's clustering objective does not itself prescribe a federated acceptance policy for those points. Likewise, the result is not a reproduction or refutation of FLAME's combined clustering, clipping, and noise mechanism. The experimentally supported distinction is between passing a consensus test and never entering a group to which that test is applied.

C. Heterogeneous Optimization and Evaluation

Unequal local computation is a separate source of heterogeneity. Wang et al. [10] analyze objective inconsistency under heterogeneous local updates and propose normalized averaging. Our fixed-step intervention controls the number of local optimizer steps while retaining the existing aggregation rule; it is not FedNova and does not test that method. The subsequent batch-regime branches address a different question: how changes in submitted updates affect acceptance at a common model checkpoint.

Poisoning evaluation also depends on adversary strength. Random perturbations, distance-constrained coordinated attacks [11], and targeted backdoors [3] test distinct failure modes. A weak perturbation that barely damages an unprotected mean cannot establish strong defense effectiveness. The present work therefore includes multiple attack families and paired controls, while limiting conclusions to their concrete implementations.

The experiments are centralized simulations with statistical heterogeneity and controlled participation. HAR supplies smartphone activity features, but the evaluation does not measure radio traffic, device energy, asynchronous deployment, or privacy leakage. Access to individual plaintext updates is assumed; secure aggregation compatibility and differential privacy are not established.

D. Evaluation Studies and the Scope of Our Contribution

The conflict between poisoning detection and legitimate minority updates predates this study. Singh et al. [12] explicitly investigate fair poisoning detection on non-IID client data. Fedward [13] combines update transformation, adaptive OPTICS clustering, and clipping for heterogeneous federated backdoor defense. Our observations therefore do not establish that minority-aware detection or adaptive density clustering is a new problem or design principle.

FLPoison [14] benchmarks 15 poisoning attacks and 17 defenses across federated algorithms and data heterogeneity. Its reported protocol includes a Dirichlet concentration of 0.5 for non-IID evaluation. Our extreme-heterogeneity block and local decision traces answer a narrower question; different protocols preclude direct numerical ranking against that benchmark. Beyond federated aggregation, Abad et al. [15] examine inconsistencies in backdoor-defense evaluation, including benign-condition reporting and hyperparameter selection. Retaining failures and reporting clean controls are evaluation practices, not by themselves a claim of methodological novelty.

Closer architectural precedents also separate grouping from acceptance. DeepSight [16] combines per-model suspicion labels with cluster-level voting; FLShield [17] validates representative models using client feedback. FLAC [18] combines coarse and fine clustering with subsequent clipping. These designs rule out treating the separation of clustering and acceptance as a new principle. Our narrower observation concerns which updates enter the additional consensus test in the local SNNC implementation. The cited descriptions are not evidence that these other methods share its treatment of unclustered updates, and we have not reproduced their noise-label handling.

The distinguishing evidence offered here is an implementation-specific trace connecting recovered-cluster membership to additional rejection at matched states. A selected low-density group can cease to undergo consensus rejection without passing that test. This is a bounded case analysis, not an established first discovery, a replacement for broader benchmarks, or a demonstrated remedy. Generalizing the pathway or using it to justify a new defense requires independent interventions and security evaluation.

III. SYSTEM AND THREAT MODEL

At round r , the server broadcasts weights w_r . An active client trains locally and submits $u_{r,i} = w_{r,i}^{\text{local}} - w_r$. These vectors are multi-step model deltas, not single-batch gradients. For an accepted set B_r , the full method uses

$$w_{r+1} = w_r + \frac{1}{|B_r|} \sum_{i \in B_r} u_{r,i}. \quad (1)$$

Uniform averaging and example-count-weighted averaging are separate baselines. Calling both simply FedAvg would conceal an experimental difference.

The server is trusted and clients have fixed identities. Attackers manipulate their own training or updates; the experiments do not implement identity creation or a general Sybil adversary. Gray-box random-direction probes do not

observe other clients' updates. In contrast, the Min-Max and Min-Sum implementations explicitly access the current benign updates and are labeled omniscient. These knowledge models are not interchangeable. Ground-truth attacker labels are used for evaluation and the explicitly marked oracle block, not supplied to the deployable full filter.

Attack ratios in the manifest refer to the configured client population. The evaluation records actual participating malicious identities per round, because conditional participation changes the realized ratio. Main runs use a fixed round-size policy with configured dropout 0.1 and an always-participate malicious policy. Paired controls retain the corresponding experimental identities and schedules while disabling attacks.

IV. LAYERED FILTERING METHOD

A. Geometric Acceptance Region

Given n submitted updates, the initial layer approximates

$$m = \arg \min_z \sum_{i=1}^n \|u_i - z\|_2 \quad (2)$$

by Weiszfeld iteration, with at most 100 iterations and tolerance 10^{-5} . It accepts

$$B_0 = \{i : \|u_i - m\|_2 \leq 2.5 \text{ median}_j \|u_j - m\|_2\}. \quad (3)$$

If the raw set in (3) contains fewer than half the submitted updates, the implementation replaces it with the full participant set; for at most two updates it also accepts all. We use B_0 below for this safeguarded set. The radius is a multiple of the median radial distance; it is not a calibrated median-absolute-deviation hypothesis test. Robustness of a location estimator does not imply robustness of this acceptance region followed by an ordinary mean. In particular, the geometric median's breakdown properties do not prove poisoning resistance or honest-client retention for the complete pipeline.

B. Density Gap and Alarm Gate

With effective neighborhood size $k' = \min(5, n - 1)$, the implementation computes

$$\text{rd}(i) = \frac{k'}{\sum_{j \in k' \text{NN}(i)} \|u_i - u_j\|_2}. \quad (4)$$

Zero-distance and small-population cases have explicit numerical branches in the frozen implementation. Consecutive gaps in sorted densities define a maximum-to-median gap ratio. A ratio above 10 triggers a density-gap indicator and places a split at the largest gap. Otherwise a tenth-percentile threshold is available when a previous alarm keeps the upper layer active.

The current-round attack gate requires both the gap indicator and at least $\max(2, \lceil 0.05n \rceil)$ initial-layer rejections. Both statistics are computed from the same updates: this is corroborating geometric evidence, not statistically independent evidence. The threshold is a heuristic; no calibrated significance level or distribution-free false-alarm bound is asserted.

C. Cluster Validation and Temporal State

Shared-neighbor clustering groups low-density updates. In the local implementation, each point retains up to k' neighbors within three times the estimated radius. Points sharing at least one retained neighbor form overlapping sets, which are merged. Only resulting sets whose size is at least the larger of two and the mean resulting-set size are designated natural clusters. Low-density points outside these clusters are not subjected to a cluster-consensus rejection; they remain eligible if they pass B_0 . When a density gap exists, the neighborhood-radius estimate excludes the low-density subset; otherwise it uses the broader population. This conditional estimation does not guarantee that the retained subset is uncontaminated.

For a recovered group S , centroid c_S , and geometric median m_{B_0} of the initial accepted pool, the group is rejected if

$$\|c_S - m_{B_0}\|_2 > 2 \max(10^{-10}, \text{median}_{j \in B_0} \|u_j - m_{B_0}\|_2). \quad (5)$$

The ordinary combined decision is $B = B_0 \setminus \bigcup_{S \text{ rejected}} S$. Thus an upper layer cannot recover a client rejected by B_0 . This set-inclusion property constrains minority recovery as well as attack rejection; it is not a security guarantee.

A three-round memory of gate decisions can sustain the alarm until the clean-streak condition suppresses it. The alarm controls whether the upper filtering layer is active. It is therefore not a side channel independent of filtering. If the combined layer accepts fewer than half the submitted updates, the valve reverts to B_0 . The initial safeguard ensures that B_0 contains at least half the submitted updates. This count guarantee does not identify honest participants or ensure useful minority retention; with attackers present it is not an honest-client FPR guarantee.

Algorithm 1 summarizes the full variant with all layers enabled. The density split and radius follow the definitions above; an empty history contributes no active gate. Numerical validation, populations of at most two updates, and ablation switches remain explicit implementation branches. Crucially, the loop removes only members of rejected natural clusters. A point in L outside every $S \in \mathcal{S}$ stays in B if it passed B_0 .

D. Execution Contract

The audit contract records accepted IDs, fallback decisions, identities, and per-round metrics. An invalid empty acceptance decision is governed by the configured fallback policy; main audit-v2 uses an explicitly degraded accept-all policy. Nonfinite updates are not silently reclassified as successful results. These operational branches must be distinguished from the ordinary set-inclusion rule and from successful attack detection. The reference implementation is frozen by source hashes; neither the recovery attempts nor the subsequent CIFAR diagnostics overwrite it.

V. EXPERIMENTAL PROTOCOL

A. Data, Models, and Training

The matrix uses MNIST [19], Fashion-MNIST [20], UCI HAR [21], and CIFAR-10. MNIST and Fashion use a 784–200–10 MLP. HAR uses a 561–128–64–6 MLP with dropout

Algorithm 1 Full-filter decision for a validated round ($n \geq 3$)**Require:** Updates $U = \{u_i\}_{i=1}^n$, gate history H , clean streak

```

 $c$ 
1:  $B_0 \leftarrow$  geometric acceptance set in (3)
2: if  $|B_0| < n/2$  then
3:    $B_0 \leftarrow \{1, \dots, n\}$ 
4: end if
5: Compute relative densities, gap indicator  $g$ , split  $t$ 
6:  $q \leftarrow g \wedge (n - |B_0| \geq \max(2, \lceil 0.05n \rceil))$ 
7:  $a \leftarrow q \vee (c < 3 \wedge \text{any}(H[-3 :]))$ 
8:  $B \leftarrow B_0$ 
9: if  $a$  then
10:   $L \leftarrow \{i : \text{rd}(i) < t\}$ ; estimate radius  $\varepsilon$ 
11:   $\mathcal{S} \leftarrow \text{SNNC}(U_L, k = 5, \varepsilon)$ 
12:   $m_0 \leftarrow \text{GeoMedian}(U_{B_0})$ 
13:   $R \leftarrow 2 \max(10^{-10}, \text{median}_{j \in B_0} \|u_j - m_0\|_2)$ 
14:  for each natural cluster  $S \in \mathcal{S}$  do
15:    if  $\|\text{mean}_{i \in S} u_i - m_0\|_2 > R$  then
16:       $B \leftarrow B \setminus S$ 
17:    end if
18:  end for
19:  if  $|B| < n/2$  then
20:     $B \leftarrow B_0$ 
21:  end if
22: end if
23: Append  $q$  to  $H$ ; set  $c \leftarrow 0$  if  $q$ , else  $c \leftarrow c + 1$ 

```

Ensure: Accepted set B , alarm a , updated temporal state

TABLE I
PLANNED AUDIT-V2 COVERAGE. A UNIT IS ONE
METHOD-CONDITION-SEED RUN; THE BLOCKS ARE NOT A BALANCED
FULL FACTORIAL.

Block	Jobs	Units
main	67	1500
clean	26	156
oracle	26	78
ablation	12	252
temporal	6	36
preserve_empty	6	54
fixed_steps	6	54

0.2. CIFAR uses two convolutional layers with 32 and 64 channels, pooling, and fully connected layers of widths 256 and 10, with dropout 0.25. Main configurations use 100 clients, 30 rounds, seeds 42, 137, and 2024, batch size 32, three local epochs, learning rate 0.01, and optimizer momentum 0.9. Optimizer momentum is distinct from the alarm-memory ablation.

Dirichlet concentration [22] varies over 0.5, 0.1, and 0.01 in the base block. Main partitioning uses a minimum-count repair policy with 20 examples per client and records the number moved. Therefore the final partition should not be described as an untouched Dirichlet draw. Preserve-empty and five-local-step blocks examine alternative partition/training policies. The fixed-step control matters because three epochs imply different optimization effort for clients with different sample counts.

The plan contains 149 jobs and 2,130 units, partitioned as in Table I. It is not a full factorial across every dataset, attack,

and method. Three original datasets share a 17-condition base block, whereas CIFAR has a smaller condition set. Supplementary CSVs and the scenario manifest enumerate the exact coverage.

B. Attacks and Controls

Loud Gaussian noise uses per-coordinate standard deviation 5. The stealth probe adds a random-direction perturbation whose total norm is 0.005 times the original update norm. The legacy identifier `adaptive_gaussian` denotes bounded random-direction noise at relative scale 0.3; it is not an optimized defense-adaptive attack, nor does it preserve the update norm exactly. Label flipping modifies local training labels.

The omniscient Min-Max and Min-Sum probes choose a common update along the negative benign-mean unit direction, with a scalar constrained by the respective benign distance bounds. They test that specified direction, not all possible optimized attack constructions. Patch backdoors use a bottom-right 3×3 white patch, target class 0, and configured poisoning fraction 0.2. Attack-disabled controls retain trigger evaluation so that naturally occurring target predictions are visible. Oracle results are diagnostic references rather than deployable methods.

C. Baselines and Ablations

The main baseline family comprises uniform mean, sample-weighted mean, norm clipping, coordinate median, Multi-Krum with $f = \lceil 0.3n \rceil$ and $m = n - f - 2$ (61 selected updates when $n = 90$), normalized FLTrust with a 100-example root set, and the HDBSCAN FLAME implementation. These labels identify the concrete local implementations; matching every original paper's training and privacy calibration is not claimed. In particular, the FLAME noise setting is 0.001, and performance here is not a universal assessment of FLAME. Six small-input checks through the local server agree with independently computed clipping, averaging, and injected-noise scaling, but do not compare clustering decisions with the author implementation. If clustering returns no accepted update, the configured degraded fallback accepts all updates and still applies FLAME clipping and noise. No such flagged fallback or all-accepted round occurs among the 6000 recorded rounds of the 200 completed canonical FLAME runs; the separate failed run remains a failure. Package-level clustering equivalence and dataset-specific noise calibration remain unverified. A limited check of normalized FLTrust compares its aggregation rule against a NumPy transcription of the author-demo rule on 18 recorded update matrices and four constructed root vectors, 72 input cases per variant. Relative aggregate discrepancy stays below 2.8×10^{-7} ; running a subset of six matrices with the four root constructions through the local server path rather than a restated formula gives 2.4×10^{-7} . This is agreement of the aggregation rule on the inputs tested, not fidelity of the whole training protocol, and the author framework itself was never executed. Two boundary cases do diverge: with a degenerate root update the local filter falls back to a uniform mean and raises no flag, and

with every trust score zero it applies its configured accept-all-degraded fallback, where the published rule would apply nothing. Recorded rounds carry neither the root magnitude nor the executed aggregation branch, so 135 of 5490 rounds in which every client was accepted cannot be attributed to one path or the other. Our global update multiplier is one; matching local and root epochs does not guarantee matching optimizer-step counts. The effect of that training-protocol difference remains unmeasured.

Seven ablation variants remove upper filtering, momentum, or the valve, or vary the trust-region multiplier. The `fed_g2l_25` identifier is a geometric-median-distance control, not evidence of reproducing the complete published FedG2L system. Each variant has 36 matched endpoints across three datasets and four conditions. Full method parameters remain fixed within the canonical evaluation.

D. Metrics and Failure Accounting

The primary utility metric is final-round test accuracy. Honest-client FPR is $FP/(FP + TN)$ and attacker rejection rate is $TP/(TP + FN)$, pooled over participating clients. For normalized FLTrust, the client FPR/TPR fields count honest/attacker updates assigned zero trust weight by the local cosine rule, rather than a separate attack-classification decision. This describes zero contribution in the ordinary weighting path; fallback behavior must be interpreted separately. These fields are not directly interchangeable with explicit rejection decisions from other defenses. Alarm FPR is instead computed over attack-free rounds. A false alarm cannot be assessed by restricting the denominator to attacked rounds. Backdoor ASR uses triggered examples excluding the target class. Balanced accuracy and client-group counts are retained in the supplementary data.

We use matched method and attack-disabled differences where the report establishes the corresponding identities. Descriptive averages weight the specified conditions and seeds equally; they do not estimate a deployment's unknown attack prevalence. Three seeds and correlated conditions do not justify treating every round or client as an independent replicate. No significance claim or legacy Wilcoxon result is carried into this revision. Failed runs remain explicit outcomes, rather than zero-accuracy imputations or silently replaced seeds.

VI. RESULTS

A. Coverage and Utility

The canonical matrix contains 2,125 valid results and five documented failures. Resource-related incomplete work was recovered separately while preserving prior successful artifacts. Four persistent failures belong to sample-weighted HAR training and one to CIFAR FLAME under Min-Max. Table II summarizes the common base block and does not support a claim that the full method has the highest mean accuracy across datasets. The conditional HAR sample-weighted entry cannot be ranked fairly against complete entries without accounting for failure coverage.

TABLE II

DESCRIPTIVE ACCURACY (PERCENT) OVER THE SHARED 17-CONDITION BLOCK, THREE SEEDS EACH. *HAR SAMPLE-WEIGHTED MEAN HAS ONLY 47/51 SUCCESSFUL UNITS; ITS MEAN IS CONDITIONAL ON SUCCESS AND NOT DIRECTLY COMPARABLE. CIFAR HAS A DIFFERENT CONDITION SET AND IS EXCLUDED.

Method	MNIST	Fashion	HAR
Full	67.90	63.16	50.32
Uniform mean	83.79	73.31	49.46
Sample-weighted	86.63	74.22	49.59*
Norm clipping	81.46	73.77	59.19
Coord. median	59.20	58.41	35.10
Multi-Krum	71.23	64.57	47.91
FLTrust norm.	83.76	74.51	44.27
FLAME HDBSCAN	59.38	56.37	41.93

TABLE III

ATTACK-FREE $\alpha = 0.01$: FINAL ACCURACY AND POOLED HONEST-CLIENT FALSE REJECTION (PERCENT); FALSE ALARMS OVER 90 CLEAN ROUNDS. THREE SEEDS PER DATASET.

Dataset	Acc.	Client FPR	Alarm FP
MNIST	26.43	42.57	90/90
Fashion	31.77	38.69	90/90
HAR	35.09	24.37	90/90
CIFAR-10	16.03	39.94	90/90

B. Attack-Free Heterogeneity

Table III isolates the extreme attack-free condition. The full method rejects 24.37–42.57% of honest submitted updates. All 90 rounds per dataset raise an alarm. We checked the twelve underlying run files: actual malicious identity lists are empty and the alarm counts agree with the report. Heterogeneity alone is sufficient to activate both the initial outlier support and the density gate in these cases.

These outcomes contradict general interpretations of the alarm as trustworthy attack attribution. They also prevent aggregate FPR from being used as evidence of minority protection. A minority-specific claim would require group-conditioned retention together with learning outcomes; the supplementary group records enable this analysis but the aggregate table alone cannot establish it.

C. Contribution of Additional Layers

Table IV reports variant-minus-full final accuracy. The full architecture gains only 0.081 percentage points on average over L0 alone. Condition-specific differences range in both directions, so the mean is not a universal equivalence statement. The no-momentum variant has exactly the same final accuracy as the full method at all 36 endpoints, verified against raw run files. This does not show that temporal alarm trajectories are identical.

Removing the valve improves accuracy by 0.278 points on average in this block, with mixed condition-specific effects. Changing the trust multiplier also has nonmonotonic outcomes. Selecting a preferred variant after observing these results would constitute further development; independent validation would be required for a resulting improved-method claim.

TABLE IV
VARIANT MINUS FULL FINAL ACCURACY (PERCENTAGE POINTS), 36
MATCHED ENDPOINTS PER VARIANT. RANGES DESCRIBE
HETEROGENEITY, NOT CONFIDENCE INTERVALS.

Variant	Mean	Min.	Max.
L0 only	-0.081	-4.61	+6.65
GM-distance control	-0.081	-4.61	+6.65
No momentum	+0.000	+0.00	+0.00
No valve	+0.278	-0.73	+8.74
RTR 1.5	-0.916	-20.42	+14.69
RTR 2.0	-0.860	-7.22	+9.90
RTR 3.0	+0.575	-20.53	+24.49

TABLE V
PATCH ATTACK (MAIN 8.1, $\alpha = 0.1$) AND PAIRED ATTACK-DISABLED
CONTROLS: FULL METHOD, PERCENT. ASR IS MEASURED ON TRIGGERED
NON-TARGET EXAMPLES.

Condition	Accuracy	ASR
MNIST	93.97	99.993
Matched clean	94.14	0.573
Fashion	83.34	99.930
Matched clean	83.42	2.630
CIFAR-10	49.55	96.589
Matched clean	49.85	5.756

D. Backdoor Success Despite Clean Utility

Under the tested moderate-heterogeneity patch attack, the full method reaches ASR of 99.993% on MNIST, 99.930% on Fashion, and 96.589% on CIFAR (Table V). Corresponding attack-disabled control ASRs are much lower. High clean test accuracy therefore masks a targeted security failure. Conversely, the weaker clean models in the extreme-heterogeneity block must not be credited as effective defenses merely because their ASR is lower.

E. Numerical Failures and CIFAR Repetition

The four HAR failures occur for sample-weighted mean in conditions 3.1 (seeds 42 and 2024) and 3.2 (seeds 137 and 2024). Isolated repetitions reproduce numerical failure, with nonfinite parameters or updates observed during local training. Correctly normalized aggregation weights do not by themselves ensure stable local optimization. These cases remain failed outcomes under the evaluated protocol.

The canonical CIFAR FLAME condition 7.1/seed 42 fails in both the original and isolated recovery attempts, while an instrumented diagnostic completes at 10% accuracy. Two subsequent three-round non-deterministic-profile repetitions already differ at the first client's first update. Their partition and initial-model hashes match. Under a deterministic GPU profile, two short runs and an instrumented short run instead yield matching traces.

Two separate 30-round deterministic-profile repetitions then complete with all 2,730 recorded trace entries matching and final accuracy 9.94% in both. The profile enables deterministic algorithms and cuDNN determinism and configures the cuBLAS workspace. This supports repeatability for this condition and environment; it does not establish portability, diagnose one specific GPU kernel, or validate every CIFAR

condition. These diagnostic runs are not inserted into the canonical five-failure matrix, because the backend profile differs. Completion and predictive robustness remain separate outcomes.

F. Exploratory Checkpoint Diagnostics

We conducted a separate attack-free study at Dirichlet concentration 0.01 with five local optimizer steps. Nine 30-round reference trajectories cover MNIST, Fashion-MNIST, and HAR with the same three seeds. At zero-based rounds 0, 9, 19, and 29, three independent one-round branches start from the same model, participating clients, partition, and defense history. Arm A retains the standard batch-32 loader; B draws a fresh 20-example subset per step, without replacement within a step; C repeats the first subset for all five steps. Branches never feed back into the reference. A/B changes both batch size and sampling regime; B/C changes subsequent example identities and order. Neither isolates only the number of unique examples.

All 108 branches completed. In all 36 checkpoint cells, A reproduces the recorded reference accuracy, accepted IDs, and update norms. The matched contrasts are heterogeneous: at MNIST seed 137, round 0, FPR is 23.33%, 15.56%, and 5.56% for A/B/C. At seed 2024, round 9, it is 37.78%, 27.78%, and 20.00%. Separate arm medians across three seeds conceal these contrasts. The study therefore supports neither universal absence of a batching effect nor universal improvement from repeated batches.

Figure 1 displays the corresponding acceptance accounting for the selected forward checkpoint. For the latter checkpoint, three instrumented repetitions reproduce the original input-update matrix hashes and final decisions exactly. A rejects 20 clients at L0 and 14 additionally; C rejects 14 at L0 and four additionally. Of the ten clients avoiding additional rejection, all remain on the low-density side but become singleton groups outside the natural clusters tested for consensus. They do not switch to passing that test. The alarm remains active and no safety fallback occurs in either arm. The supplementary evidence reports cluster membership and neighborhood thresholds. This checkpoint was selected after inspecting the branch results, so it is an explanatory case rather than independent validation.

G. Decision Pathway and Exploratory Detection Signals

Matched fixed-matrix replays isolate why the implemented filter does not reject the coordinated attackers it faces. Across 72 checkpoint matrices and both neighborhood-cutoff settings, opening the gate on the density indicator alone left the accepted set unchanged in all 144 comparisons. The two attacks resist for different reasons. Under Min-Max the initial layer accepts every update and all updates lie within twice the median consensus radius; the arithmetic centroid of any subset of a Euclidean ball remains inside that ball, so regrouping clusters cannot by itself produce a consensus rejection. Under the patch backdoor the density gate does open, in 17 of 18 attacked checkpoints, and most attacker observations are low-density: 285 of 306 in the processed partitions. With the

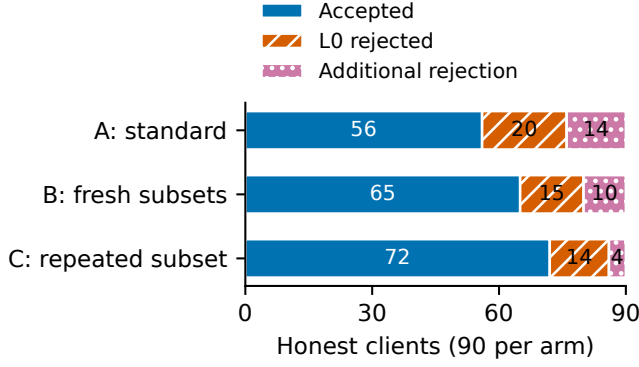


Fig. 1. Selected MNIST/2024/round9 checkpoint: partition of the same 90 honest clients under each batch regime. Additional rejection means passing L0 but failing the final filter. This is a descriptive selected case; the full paired contrasts are supplied separately.

cutoff enabled 239 of them enter natural clusters, and 30 of the 55 recovered clusters contain attackers. No recovered cluster is rejected by the consensus test. These are fixed-matrix outcomes at recorded model and momentum states; they are not equivalent to training with a modified gate and they carry no attack-success measurement.

A further fixed-geometry intervention tests the treatment of unclustered low-density updates. We apply the unchanged consensus test to each such L0-accepted update individually, while preserving the natural clusters and safety valve. Across the same 72 matrices, this adds no attacker rejections; final honest rejections increase by 32 under the original gate and by 77 under the density-only gate. These are client-checkpoint counts from the same development inputs, not independent validation. In the historical A/B/C case, rejection counts become 44/44/44 instead of 34/25/18; all ten clients previously avoiding additional rejection in C fail its singleton test. Directly excluding the unclustered candidates instead activates the safety valve in all three historical arms, returning rejection counts to the L0 values of 20/15/14. The supplementary policy table reports the broader controls. Extending test coverage therefore explains decisions here without demonstrating a security improvement or measuring accuracy or attack success.

Because the recorded geometry does not separate these attackers, we examined two directional signals that the architecture does not currently use. The first compares an update with its neighbors. On the attacked Min-Max checkpoints the mean cosine to the five nearest updates separates attackers perfectly. The implementation copies one poisoned vector to every attacker, yet constraint-valid perturbations that break exact duplication still preserve AUC 1, so the ranking detects coordinated directions without establishing malicious intent. On the patch attack the duplicate indicator is uninformative at every checkpoint and unlabeled direction scores average 0.413 and 0.443, below chance. Synthetic controls show the rule is not specific: eighteen honest clients sharing one direction among seventy-two orthogonal honest clients are all rejected, while one to five attackers on the same orthogonal background all escape.

The second signal assumes a clean server-side root set. Using one hundred held-out examples, verified disjoint from every client pool after minimum-count repair, we recorded at each of 72 checkpoints both the increase in root loss caused by a client update and the angle between that update and the negative root gradient. The loss score is rank-correlated 0.82 with the update norm and ranks attackers below honest clients in every condition; these observations do not support interpreting it as an attack-specific score. The angular score has pooled rank correlations of -0.17 with update norm and -0.12 with client sample count; these correlations do not establish independence. In the unrotated Min-Max sweep, median checkpoint AUC is 1.00 on Fashion-MNIST and 0.833 on MNIST, with checkpoint minima of 0.681 and 0.167. When each attacker independently draws a direction at a fixed angle from the inverse-mean direction and the same Min-Max distance constraint is re-solved, median AUC reaches 0.512 and 0.444 at ninety degrees, with no recorded constraint violation. Damage from this rotated construction was not measured. A threshold on within-checkpoint standardized scores, selected to retain all Min-Max attackers in development data, flags 64.69% of honest client-round observations in the corresponding attack-free checkpoints. Applied to attacked patch checkpoints, it flags 61.57% of honest observations and 70.37% of attacker observations. These pooled descriptive rates do not establish statistical equivalence or rule out other operating points.

A separate exploratory training experiment examines the Spearman trend of each client's update norm relative to the round median over 30 rounds. It uses MNIST and Fashion-MNIST, three seeds, uniform FedAvg and the full method, five local steps, $\alpha = 0.01$, and 20% attackers. An omniscient Min-Max variant targets the honest median norm or twice that norm, giving 24 completed runs and 720 rounds with no recorded distance-constraint violation. Its formula was corrected after an invalid initial run; this stage is not fully preregistered. For target ratio one, median trend AUC falls from 0.994 to 0.585. Median final-accuracy loss relative to matched clean runs changes from 0.4526 to 0.3376 for FedAvg and from 0.1830 to 0.13525 for the full method. The ratios of these medians are 74.6% and 73.9%; they are not medians of paired damage fractions. Positive-baseline paired fractions range from 2.48% to 94.68%, and one full-method baseline has negative loss, making a retained-damage percentage inappropriate. These results concern this offline scalar score; no online detector or backdoor protection was tested.

VII. DISCUSSION AND LIMITATIONS

The central limitation is a mismatch between geometric structure and adversarial intent. Honest minority updates can lie outside the initial region; later cluster validation cannot recover them because its ordinary acceptance set is a subset of B_0 . Conversely, a backdoor can retain useful clean-task behavior and evade rejection. Adding temporal memory can prolong an alarm without supplying new evidence of maliciousness.

Accordingly, neither the set-inclusion property nor the use of a robust center establishes an end-to-end security theorem.

TABLE VI
SCOPE OF THE EXPLORATORY CHECKS. DISTINCT INTERVENTIONS SUPPORT DISTINCT CONCLUSIONS; THEY DO NOT ESTABLISH A COMMON FAILURE MECHANISM OR EXHAUST UPDATE GEOMETRY.

Component or score	Intervention	Observation	Damage evidence and scope
Initial acceptance region	Attack-free heterogeneity; matched recorded attack matrices	Honest rejection in clean runs; Min-Max and patch updates accepted in the diagnostic block	Canonical learning and ASR measured separately; replay is not modified-filter training
Neighbor-direction score	Constraint-valid perturbations; synthetic honest coalitions	AUC 1 survives small perturbations; aligned honest groups can be rejected	No damage measurement for synthetic controls; a specificity counterexample
Clean-root angular score	Independent directions at fixed angles, re-solving the distance constraint	Median AUC at 90°: 0.512 / 0.444	Rotated-update damage unmeasured; detectability only
Relative-norm trend	Profile-aware Min-Max training, target ratios one and two	At ratio one, median AUC 0.994 → 0.585	Final-accuracy loss measured; median ratios 74.6% / 73.9%; exploratory, no ASR

The gap test needs calibration on attack-free heterogeneous populations before operational alert precision can be claimed. Filter decisions, clean utility, alarm specificity, and targeted ASR require separate evaluation. There is no demonstrated system-level privacy, communication-efficiency, or IoT-energy benefit in this simulation.

The checkpoint diagnosis exposes a dependence on whether a low-density update belongs to a recovered group: an isolated update can remain accepted while a coherent group fails consensus. This behavior has been traced in the local implementation, not established for an entire family of clustering defenses. In the batch-regime branches, changes in the shared-neighbor graph, radius, and initial acceptance pool occur together, so their individual contributions remain unresolved. The subsequent fixed-geometry policy intervention isolates test coverage at those recorded inputs: singleton validation adds honest rejections without additional attacker rejections in the tested attack block. It does not demonstrate an adversarial exploit or improved learning. Client sample count, label composition, and minimum-count repair are also not separated. First-mini-batch losses do not establish whole-client optimization difficulty.

Table VI separates the evidence supporting each limitation. Honest rejection, lack of specificity in synthetic coalitions, direction-sensitive root scores, and attenuation of one temporal norm score are different observations. Only the profile-aware temporal intervention measures the resulting training damage; the rotated-direction construction does not. At ninety degrees its added component is orthogonal to the honest mean, while the full update has dot product $\|\text{mean}\|^2$ with that mean. Constraint feasibility alone therefore does not establish preserved attack damage. These results constrain the tested rules and scores, without establishing that all geometric or temporal defenses fail or that new information is necessary for every improvement.

The evidence is limited by three seeds, short training horizons, small models, and uneven dataset coverage. Canonical CIFAR results are descriptive observations under a backend profile with demonstrated repeatability limitations. Baseline fidelity, attack direction choices, data-repair policy, and conditioning on successful outcomes also bound interpretation. An improved filter should be developed on designated conditions and tested on held-out seeds and attack settings; rerunning the

same matrix until favorable outcomes occur would not address these limitations.

VIII. CONCLUSION

The evaluated Fed-MDBSCAN-G implementation exhibits substantial honest-client rejection and false alarms under attack-free extreme heterogeneity, a small mean accuracy contribution from its upper layers in the paired ablation block, and high attack success in the canonical patch-backdoor block. These results concern the recorded protocols and retain the five failed runs; they do not rank the full method as generally superior or establish failure of geometric defenses as a class.

The selected shared-checkpoint contrast identifies a concrete acceptance pathway: low-density updates can avoid additional rejection by remaining outside natural clusters tested for consensus. Fixed-matrix attack replays expose distinct pathways for Min-Max and patch updates. Extending consensus testing to unclustered candidates adds no attacker rejections in that block and increases honest rejection. These diagnostics explain local decisions without demonstrating that changing them improves long-horizon learning or security.

The candidate-score studies also have distinct limits. Aligned honest groups undermine neighbor-score specificity; a direction sweep weakens the clean-root score without measuring the rotated construction's damage; profile-aware training weakens one offline temporal score while retaining substantial median accuracy damage, with heterogeneous paired outcomes. Further defense development requires separately specified interventions and held-out evaluation. The present contribution is the measured behavior and traced decision pathways of this implementation, bounded by exploratory selection, baseline fidelity limits, and the recorded training conditions.

ARTIFACT AVAILABILITY

The accompanying evidence package contains per-unit outcomes, paired differences, group counts, configuration coverage, and source/report hashes. Raw training artifacts and the frozen implementation are retained in the project archive. No public repository URL, archival DOI, or third-party accessibility is asserted until a release is actually made.

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